ECE590 Homework 2

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February 2026

1 Question 1

Setup. We are given a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

with $y_i \in \mathbb{R}$ and $x_i \in \mathbb{R}^d$. The Gaussian conditional model is

$$p_\theta(y_i \mid x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f_\theta(x_i))^2}{2\sigma^2}\right), \quad f_\theta(x) = w^\top \psi(x) + b,$$

where $\theta = (w, b)$ and $\psi : \mathbb{R}^d \to \mathbb{R}^{d'}$ is a nonlinear feature map.

(a) Negative log-likelihood and squared loss. Assume $\sigma^2 = 1$. For i.i.d. data,

$$p_\theta(\mathcal{D}) = \prod_{i=1}^N p_\theta(y_i \mid x_i).$$

The negative log-likelihood is

$$-\log p_\theta(\mathcal{D}) = \sum_{i=1}^N \left[\frac{1}{2}\log(2\pi) + \frac{1}{2}\big(y_i - w^\top \psi(x_i) - b\big)^2\right].$$

The constant term $\frac{1}{2}\log(2\pi)$ does not depend on θ, hence minimizing the negative log-likelihood is equivalent to minimizing

$$\ell_\theta(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^N \big(y_i - w^\top \psi(x_i) - b\big)^2.$$

(b) Gradients with respect to b and w. Define the residual

$$r_i = y_i - w^\top \psi(x_i) - b.$$

Then

$$\ell_\theta(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^N r_i^2.$$

Derivative with respect to b:

$$\frac{\partial \ell_\theta(\mathcal{D})}{\partial b} = -\frac{2}{N}\sum_{i=1}^{N} r_i = \frac{2}{N}\sum_{i=1}^{N}\big(w^\top \psi(x_i) + b - y_i\big).$$

Gradient with respect to w:

$$\nabla_w \ell_\theta(\mathcal{D}) = -\frac{2}{N}\sum_{i=1}^{N} r_i\, \psi(x_i) = \frac{2}{N}\sum_{i=1}^{N}\big(w^\top \psi(x_i) + b - y_i\big)\psi(x_i).$$

(c) Input-dependent variance. Assume now that the variance depends on x:

$$\sigma_i^2 = f_\phi(x_i) = v^\top \psi(x_i) + c, \quad \phi = (v, c).$$

The negative log-likelihood becomes

$$\ell(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{2}\log(2\pi\sigma_i^2) + \frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right], \quad \mu_i = w^\top \psi(x_i) + b.$$

Ignoring constants independent of ϕ, define $r_i = y_i - \mu_i$ and write

$$\ell_i = \frac{1}{2}\log(\sigma_i^2) + \frac{r_i^2}{2\sigma_i^2}.$$

Then

$$\frac{\partial \ell_i}{\partial \sigma_i^2} = \frac{1}{2}\left(\frac{1}{\sigma_i^2} - \frac{r_i^2}{\sigma_i^4}\right).$$

Derivative with respect to c:

$$\frac{\partial \ell(\mathcal{D})}{\partial c} = \frac{1}{2N}\sum_{i=1}^{N}\left(\frac{1}{\sigma_i^2} - \frac{(y_i - \mu_i)^2}{\sigma_i^4}\right).$$

Gradient with respect to v:

$$\nabla_v \ell(\mathcal{D}) = \frac{1}{2N}\sum_{i=1}^{N}\left(\frac{1}{\sigma_i^2} - \frac{(y_i - \mu_i)^2}{\sigma_i^4}\right)\psi(x_i), \quad \sigma_i^2 = v^\top \psi(x_i) + c.$$

2 Question 2 (Logistic regression): Gradients w.r.t. b and w.

We consider $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^d$. The logistic regression model is

$$p_\theta(y_i \mid x_i) = \big[\sigma(z_i)\big]^{y_i}\big[1 - \sigma(z_i)\big]^{1-y_i}, \qquad z_i = w^\top \psi(x_i) + b,$$

where $\theta = (w, b)$ and $\sigma(z) = \frac{1}{1+e^{-z}}$.

The average negative log-likelihood is

$$\ell_\theta(\mathcal{D}) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(y_i \mid x_i) = \frac{1}{N}\sum_{i=1}^{N} \left[-y_i \log \sigma(z_i) - (1-y_i)\log\big(1-\sigma(z_i)\big)\right].$$

Let $p_i = \sigma(z_i)$. For the per-sample loss

$$\ell_i = -y_i \log p_i - (1-y_i)\log(1-p_i),$$

we use $\sigma'(z) = \sigma(z)\big(1-\sigma(z)\big) = p_i(1-p_i)$ to obtain

$$\frac{\partial \ell_i}{\partial z_i} = \left(-\frac{y_i}{p_i} + \frac{1-y_i}{1-p_i}\right) p_i(1-p_i) = p_i - y_i = \sigma(z_i) - y_i.$$

Since $z_i = w^\top \psi(x_i) + b$, we have $\frac{\partial z_i}{\partial b} = 1$ and $\nabla_w z_i = \psi(x_i)$. Therefore,

$$\frac{\partial \ell_\theta(\mathcal{D})}{\partial b} = \frac{1}{N}\sum_{i=1}^{N} \frac{\partial \ell_i}{\partial z_i}\frac{\partial z_i}{\partial b} = \frac{1}{N}\sum_{i=1}^{N}\big(\sigma(z_i) - y_i\big),$$

and

$$\nabla_w \ell_\theta(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^{N} \frac{\partial \ell_i}{\partial z_i}\nabla_w z_i = \frac{1}{N}\sum_{i=1}^{N}\big(\sigma(z_i) - y_i\big)\psi(x_i).$$

Equivalently, substituting $z_i = w^\top \psi(x_i) + b$,

$$\boxed{\frac{\partial \ell_\theta(\mathcal{D})}{\partial b} = \frac{1}{N}\sum_{i=1}^{N}\left(\sigma\big(w^\top \psi(x_i) + b\big) - y_i\right), \qquad \nabla_w \ell_\theta(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^{N}\left(\sigma\big(w^\top \psi(x_i) + b\big) - y_i\right)\psi(x_i).}$$

3 Question 3 (Softmax model): Gradients w.r.t. (A, b) and an equivalent parameterization with $w_C = \mathbf{0}_m$.

Setup. We have labels $y_i \in \{1, \dots, C\}$ with $C > 2$ and inputs $x_i \in \mathbb{R}^d$. Let the feature map be $\psi(x) \in \mathbb{R}^{d'}$ and define the latent vector

$$f(x) = A\psi(x) + b \in \mathbb{R}^m, \qquad \theta = (A, b),$$

where $A \in \mathbb{R}^{m \times d'}$ and $b \in \mathbb{R}^m$. The class vectors $w_1, \dots, w_C \in \mathbb{R}^m$ are *known and fixed*. For sample i, denote

$$f_i \triangleq f(x_i) = A\psi(x_i) + b, \qquad s_{ic} \triangleq w_c^\top f_i,$$

and the softmax probability

$$p_{ic} \triangleq p_\theta(Y = c \mid X = x_i) = \frac{\exp(s_{ic})}{\sum_{k=1}^{C}\exp(s_{ik})}.$$

The (average) negative log-likelihood is

$$\ell_\theta(\mathcal{D}) = -\frac{1}{N}\sum_{i=1}^{N} \log p_{i\,y_i}.$$

(a) Gradient of $\ell_\theta(\mathcal{D})$ with respect to (A, b)

Step 1: Expand the per-sample loss. For a fixed i,

$$-\log p_{i\,y_i} = -\log\left(\frac{\exp(s_{i\,y_i})}{\sum_{k=1}^{C}\exp(s_{ik})}\right) = -s_{i\,y_i} + \log\left(\sum_{k=1}^{C}\exp(s_{ik})\right).$$

Hence,

$$\ell_\theta(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^{N}\left[-s_{i\,y_i} + \log\left(\sum_{k=1}^{C}\exp(s_{ik})\right)\right].$$

Step 2: Compute $\frac{\partial}{\partial s_{ic}}(-\log p_{i\,y_i})$. Let $y_{ic} \triangleq \mathbf{1}\{y_i = c\}$. Differentiate the expression in Step 1:

$$\frac{\partial}{\partial s_{ic}}\Big(-s_{i\,y_i}\Big) = -\mathbf{1}\{y_i = c\} = -y_{ic},$$

and

$$\frac{\partial}{\partial s_{ic}}\log\left(\sum_{k=1}^{C}e^{s_{ik}}\right) = \frac{e^{s_{ic}}}{\sum_{k=1}^{C}e^{s_{ik}}} = p_{ic}.$$

Therefore,

$$\boxed{\frac{\partial}{\partial s_{ic}}\big(-\log p_{i\,y_i}\big) = p_{ic} - y_{ic}}$$

Step 3: Chain rule to obtain $\nabla_{f_i}(-\log p_{i\,y_i})$. Recall $s_{ic} = w_c^\top f_i$, so $\nabla_{f_i} s_{ic} = w_c$. By the chain rule,

$$\nabla_{f_i}\big(-\log p_{i\,y_i}\big) = \sum_{c=1}^{C}\frac{\partial(-\log p_{i\,y_i})}{\partial s_{ic}}\nabla_{f_i} s_{ic} = \sum_{c=1}^{C}(p_{ic} - y_{ic})w_c.$$

Define the vector

$$g_i \triangleq \sum_{c=1}^{C}(p_{ic} - y_{ic})\,w_c \in \mathbb{R}^m.$$

Then

$$\nabla_{f_i}\big(-\log p_{i\,y_i}\big) = g_i.$$

Since $\ell_\theta(\mathcal{D})$ is the average over i,

$$\nabla_{f_i}\ell_\theta(\mathcal{D}) = \frac{1}{N}g_i.$$

Step 4: Chain rule from $f_i = A\psi(x_i) + b$ to (A, b). We have

$$\frac{\partial f_i}{\partial b} = I_m, \qquad \frac{\partial f_i}{\partial A} \text{ acts as an outer product with } \psi(x_i).$$

Thus,

$$\nabla_b \ell_\theta(\mathcal{D}) = \sum_{i=1}^{N} \nabla_{f_i} \ell_\theta(\mathcal{D}) \cdot \frac{\partial f_i}{\partial b} = \frac{1}{N} \sum_{i=1}^{N} g_i,$$

and

$$\nabla_A \ell_\theta(\mathcal{D}) = \sum_{i=1}^{N} \nabla_{f_i} \ell_\theta(\mathcal{D}) \cdot \frac{\partial f_i}{\partial A} = \frac{1}{N} \sum_{i=1}^{N} g_i \, \psi(x_i)^\top.$$

Therefore,

$$\boxed{\nabla_b \ell_\theta(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^{N} \left[\sum_{c=1}^{C} (p_{ic} - y_{ic}) w_c \right], \qquad \nabla_A \ell_\theta(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^{N} \left[\sum_{c=1}^{C} (p_{ic} - y_{ic}) w_c \right] \psi(x_i)^\top.}$$

Row/coordinate form (as requested). Let $A_{j:}$ denote row j of A and b_j the j-th component of b. Let g_{ij} be the j-th component of g_i. Then

$$\boxed{\frac{\partial \ell_\theta(\mathcal{D})}{\partial b_j} = \frac{1}{N} \sum_{i=1}^{N} g_{ij}, \qquad \nabla_{A_{j:}} \ell_\theta(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^{N} g_{ij} \, \psi(x_i)^\top.}$$

(b) Equivalent model with $w_C = \mathbf{0}_m$

Step 1: Show softmax is invariant to adding the same vector to all class weights. Let $a \in \mathbb{R}^m$ be any fixed vector and define $w'_c = w_c + a$ for all c. For any x with $f = f(x)$,

$$p'(Y = c \mid X = x) = \frac{\exp\left((w'_c)^\top f\right)}{\sum_{k=1}^{C} \exp\left((w'_k)^\top f\right)} = \frac{\exp\left((w_c + a)^\top f\right)}{\sum_{k=1}^{C} \exp\left((w_k + a)^\top f\right)}.$$

Factor out $\exp(a^\top f)$ from numerator and denominator:

$$p'(Y = c \mid X = x) = \frac{\exp(a^\top f) \exp(w_c^\top f)}{\exp(a^\top f) \sum_{k=1}^{C} \exp(w_k^\top f)} = \frac{\exp(w_c^\top f)}{\sum_{k=1}^{C} \exp(w_k^\top f)} = p(Y = c \mid X = x).$$

Hence the model probabilities are unchanged.

Step 2: Choose a shift that forces the last class vector to be zero. Take $a = -w_C$. Then

$$w'_C = w_C - w_C = \mathbf{0}_m, \qquad w'_c = w_c - w_C \quad (c = 1, \ldots, C-1),$$

and by Step 1 the resulting softmax probabilities are identical to the original model. Therefore, there exists an equivalent parameterization in which

$$\boxed{w_C = \mathbf{0}_m}$$

(with all probabilities unchanged).